

Ziyu Deng

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EDUCATION:

Virginia Tech	Blacksburg, VA, USA
PhD Student in Geosciences	Aug, 2025 – Present
- GPA: 4.0/4.0 - Advisor: Dr. Ying Zhou	
Institute of Geophysics, China Earthquake Administration	Beijing, China
Master of Science in Geophysics	Sep, 2021- June, 2025
- GPA: 3.7/4.0 - Advisor Dr. Ruiqing Zhang	
Yangtze University	Wuhan, China
Bachelor of Science in Geophysics	Sep, 2017- Jul, 2021
GPA: 3.86/4.0 (Rank: 1/49)	

RESEARCH EXPERIENCE:

Graduate Research Assistant	Virginia Tech, Aug, 2025– Present
- Research Topic: Constraining Seismic Heterogeneities in the Core-Mantle Boundary (CMB) Region. - Technical Skills: Performed high-precision measurement of ScS-S traveltimes using frequency-domain spectral division techniques. - Key Finding: Model S40RTS explains the S and ScS traveltimes and the polarity of ScS-S but not the magnitude.	
Graduate Research Assistant	Institute of Geophysics, CEA, Aug, 2023– June, 2025
- Research Topic: Mantle Transition Zone (MTZ) structure beneath the North China Craton (NCC) - Technical Skills: Processed dense seismic array data to compute high-resolution P-wave receiver functions. - Key Findings: Imaged the MTZ discontinuities, providing new constraints on the geometry and stagnation of the subducting Pacific slab.	
Graduate Research Assistant	Institute of Geophysics, CEA, Sep, 2022– Jul, 2023
- Research Topic: Crustal structural analysis in Northeast China. - Technical Skills: Compiled regional seismic datasets and applied P-wave receiver function analysis to characterize crustal velocity discontinuities. - Key Findings: Conducted a comprehensive literature review to interpret seismic results within the framework of regional tectonic evolution.	

PUBLICATIONS & PRESENTATIONS

Deng, Z., Zhou Y. (2026). Constraining Seismic Heterogeneities in the Core–Mantle Boundary (CMB) Region Using S and ScS Traveltimes. Oral presentation at the Geosciences Student Research Symposium (GSRS), Virginia Tech, Blacksburg, VA.

Deng, Z., Chen, C., Zhang, R.*, & Qiang, Z. (2025). Spatiotemporal evolution of earthquake relocations and b-value characteristics in the Weiyuan area, southern Sichuan Basin. *Seismology and Geology*. (accepted)

TEACHING EXPERIENCE

Graduate Teaching Assistant

Virginia Tech

GEOS 1034: Earth's Natural Hazards | **Fall 2025**

GEOS 1034: Earth's Natural Hazards | **Spring 2026**

SERVICE & OUTREACH

- Volunteer, Invasive Plant Removal and Ecological Restoration (Blacksburg, VA): Participated in community efforts to remove invasive species (English Ivy and Garlic Mustard) in local parks.
- Volunteer, Community Tree Planting Initiative (Blacksburg, VA): Participated in a reforestation effort to improve urban canopy and local air quality, planting native trees.
- Applied for the Department Research Award, submitted in January, 2026.
- Applied for the Department Summer Award, submitted in January, 2026.

SKILLS

- Programming: Shell and MATLAB, C and Python language
- Seismology Software: Seismic analysis, GMT, SAC and TauP.
- Languages: Chinese (Native), English (Fluent; TOEFL iBT: 104).